Numerical solution of the SIR epidemic model using Euler and Runge-Kutta Methods

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Abstract:

Ordinary Differential Equations (ODEs) are used to model dynamic processes in many disciplines including engineering, physics, and biology. In epidemiology, systems of ODEs are commonly employed to simulate the spread of infectious diseases within a population. However, many realistic models are non-linear and do not admit simple analytical solutions, making numerical methods essential for approximating their solutions. Among the most widely used approaches are the Runge-Kutta (RK) methods, which offer improved accuracy compared to simpler methods such as Euler's method.

This paper investigates the performance of several numerical methods for solving ordinary differential equations, with particular focus on Euler's method, the second-order Runge-Kutta method (RK2) and the fourth-order Runge-Kutta method (RK4). The methods are first applied to a simple ODE with a known analytical solution to evaluate convergence behaviour and global error as a function of step size. They are then applied to the non-linear Susceptible-Infected-Recovered (SIR) epidemic model to simulate the spread of disease within a population. The numerical solutions are evaluated in terms of accuracy, stability, and computational efficiency, providing insight into the relative performance of each method when modelling complex real-world systems. The results highlight the trade-offs between numerical accuracy and computational cost, illustrating the advantages of higher order Runge-Kutta methods for modelling non-linear dynamics systems.